

MARFA, TX, USA

WMO: 722640

Lat: 30.371N

Lon: 104.017W

Elev: 1478

StdP: 84.78

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93035

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.5	(c) -6.9	(d) -18.8	(e) 0.8	(f) 0.0	(g) -17.3	(h) 1.0	(i) 1.9	(j) 14.6	(k) 12.7	(l) 12.8	(m) 11.6	(n) 2.3	(o) 350	(p) 0.528

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 17.1	(c) 34.7	(d) 16.0	(e) 33.0	(f) 16.2	(g) 32.3	(h) 16.2	(i) 19.7	(j) 26.9	(k) 19.2	(l) 26.6	(m) 18.8	(n) 26.3	(o) 4.7	(p) 260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	15.3	21.1	17.4	14.9	20.9	17.0	14.6	20.9	63.5	27.1	61.6	26.8	59.8	25.9	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 12.0	(b) 10.6	(c) 9.1	(e) -12.8	(f) 37.0	(g) 2.2	(h) 1.3	(i) -14.5	(j) 37.9	(k) -15.8	(l) 38.7	(m) -17.0	(n) 39.4	(o) -18.7	(p) 40.4		
(a) 12.0	(b) 10.6	(c) 9.1	(e) -13.6	(f) 20.7	(g) 2.1	(h) 0.8	(i) -15.2	(j) 21.3	(k) -16.4	(l) 21.7	(m) -17.6	(n) 22.1	(o) -19.2	(p) 22.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.1	5.6	8.0	11.2	15.0	19.5	23.5	23.2	22.8	20.2	15.6	10.1	5.9		
	DBStd	7.30	3.83	4.35	3.89	3.61	3.20	2.06	1.78	1.81	2.44	3.56	3.84	3.91		
	HDD10.0	447	142	79	34	6	1	0	0	0	0	5	44	135		
	HDD18.3	1779	396	289	222	107	24	1	1	1	11	96	247	386		
	CDD10.0	2304	5	24	71	157	296	404	411	397	307	179	47	7		
	CDD18.3	595	0	0	1	8	61	154	153	139	68	11	0	0		
	CDH23.3	8231	1	17	107	422	1289	2129	1668	1434	809	325	29	0		
	CDH26.7	3364	0	1	11	94	546	1094	726	576	252	62	1	0		
(14) Wind	WSAvg	3.8	3.9	4.3	4.4	4.7	4.4	3.7	3.4	3.2	3.3	3.5	3.6	3.7		
(15) Precipitation	PrecAvg	365	14	14	10	13	21	45	72	59	50	32	15	14		
	PrecMax	713	46	52	53	48	85	152	152	143	128	99	117	58		
	PrecMin	235	1	0	1	1	1	8	29	21	2	1	0	0		
	PrecStd	120	15	14	11	14	20	32	34	27	38	31	25	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.4	25.7	27.9	31.1	35.0	37.6	35.1	33.9	32.7	30.1	26.2	22.5		
		MCWB	9.2	10.4	11.5	13.2	14.9	16.3	16.5	17.5	16.7	15.4	12.3	9.5		
	2%	DB	20.2	22.9	26.2	28.9	32.9	36.0	33.0	32.5	31.2	28.0	23.9	20.3		
		MCWB	8.1	9.0	10.5	12.2	14.3	15.9	17.1	17.4	16.6	14.4	11.2	9.0		
	5%	DB	18.0	21.3	24.0	27.5	31.9	33.9	32.3	31.3	29.1	27.1	22.4	18.0		
		MCWB	7.3	8.5	9.8	11.6	13.9	15.9	17.2	17.3	16.2	14.0	10.7	7.6		
	10%	DB	16.2	19.0	22.4	26.2	29.9	32.5	31.1	30.1	27.8	25.1	20.2	16.1		
		MCWB	6.4	7.7	9.2	10.9	13.4	15.8	17.2	17.4	16.2	13.4	10.0	6.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.1	12.0	14.0	15.7	18.2	20.1	20.5	20.3	20.0	18.9	14.7	11.5		
		MCDB	18.2	21.3	21.0	24.1	26.4	28.0	27.6	27.5	26.0	25.9	20.6	18.9		
	2%	WB	9.4	10.3	12.3	14.4	17.1	19.1	19.7	19.6	19.1	17.5	13.1	9.9		
		MCDB	17.7	19.9	21.6	22.6	25.6	27.4	26.8	27.0	25.3	24.1	20.6	17.8		
	5%	WB	8.0	9.3	11.0	13.1	16.1	18.5	19.1	19.1	18.5	16.3	11.9	8.5		
		MCDB	16.6	19.4	21.7	23.4	25.2	27.0	26.5	26.6	24.7	22.1	19.5	16.7		
	10%	WB	6.7	8.2	10.0	12.0	15.3	18.0	18.6	18.6	17.8	15.2	10.6	7.1		
		MCDB	15.6	18.2	20.8	23.1	24.7	26.7	26.2	26.2	24.2	20.9	18.9	15.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.8	18.1	18.8	19.0	18.4	17.1	14.7	14.2	14.6	16.6	17.4	17.2		
		MCDBR	21.6	21.9	22.2	21.4	21.2	20.0	17.3	16.7	17.4	19.2	21.0	20.8		
		MCWBR	12.3	11.7	11.1	10.4	9.1	7.3	5.3	5.4	6.4	8.7	10.9	11.8		
	5% WB	MCDBR	19.5	19.5	19.8	18.9	18.3	16.0	13.8	13.5	13.2	14.8	17.6	18.4		
		MCWBR	11.5	10.7	10.1	9.3	8.0	5.6	4.7	4.7	5.2	7.1	9.8	10.9		
(40) Clear-Sky Solar Irradiance	taub		0.245	0.251	0.264	0.295	0.311	0.339	0.364	0.344	0.325	0.282	0.261	0.247		
	taud		2.694	2.664	2.614	2.501	2.493	2.472	2.438	2.515	2.526	2.658	2.650	2.670		
	Ebn at Noon		1015	1031	1030	997	974	941	916	936	948	981	985	996		
	Edh at Noon		72	82	92	108	109	111	115	105	100	82	75	71		
(44) All-Sky Solar Radiation	RadAvg		3.76	4.75	5.98	7.07	7.52	7.40	6.69	6.32	5.63	4.91	4.00	3.48		
	RadStd		0.34	0.39	0.33	0.40	0.38	0.34	0.40	0.38	0.35	0.42	0.34	0.25		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+103

Nomenclature: See separate page